

CHAPPIDI RAMA PRAKASH

Full Stack Developer | Backend Engineer | Cloud & Real-Time Systems

+91 9390833979 | chappidiramaprakash167@gmail.com | github.com/ram294167 | Amalapuram, Andhra Pradesh

Portfolio: portfolio-eight-irid-6hjhyksnoy.vercel.app

PROFESSIONAL SUMMARY

Results-driven Full Stack / Backend Developer with hands-on experience designing and deploying scalable, production-grade systems. Proficient in building RESTful APIs, real-time communication platforms, and cloud-hosted services using Node.js, Express.js, PostgreSQL, Redis, Socket.IO, and AWS. Demonstrated ability to architect end-to-end solutions — from database schema design to EC2 deployment — for service booking, live tracking, and subscription-based products. Passionate about clean architecture, performance optimization, and delivering reliable systems at scale.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python

Backend: Node.js, Express.js, REST APIs, MVC Architecture, Role-Based Access Control (RBAC), JWT Authentication, Middleware Design

Real-Time: Socket.IO, Live GPS Tracking, Bidirectional Communication, Event-Driven Architecture

Databases: PostgreSQL, Redis — Schema Design, Query Optimization, Indexing, Caching, Session Management, Stored Procedures, Triggers

Cloud & DevOps: AWS EC2 (instance setup, SSH, security groups), AWS S3 (media storage, presigned URLs), Nginx (reverse proxy, SSL), PM2 (process management)

Tools: Git, GitHub, Postman, Linux CLI, VS Code

Integrations: Payment Gateway APIs, Third-Party Service APIs, Push Notifications, Webhook Handlers

PROFESSIONAL EXPERIENCE

Database Intern • Algonox Pvt. Ltd. Oct 2024 – Jan 2025

Database Management, Migration & Backend Integration

- Executed a full-scale data migration from Microsoft SQL Server (MSSQL) to PostgreSQL, covering schema translation, data type mapping, and post-migration validation — ensuring zero data loss and consistent query performance across both systems.
- Designed and authored SQL scripts, stored procedures, and triggers to automate repetitive data operations, reducing manual overhead for backend teams by standardizing data workflows.
- Participated in schema normalization and relational database design to support scalable, high-performance backend services — applying principles of 3NF to minimize redundancy.
- Collaborated closely with backend developers to integrate optimized database layers with RESTful API endpoints, improving response times and system maintainability.
- Conducted performance profiling of slow-running queries using EXPLAIN ANALYZE and applied indexing strategies to improve query execution plans.

PROJECTS

CoreFix MEP Services – Service Management Platform Jan 2026 – Present

Tech Stack: Node.js, Express.js, PostgreSQL, AWS EC2, AWS S3, JWT, REST APIs

- Architected and developed a multi-module backend system for a Mechanical, Electrical & Plumbing (MEP) service company — handling service booking, AMC subscription management, payment workflows, user roles, and technician job assignment.
- Designed normalized PostgreSQL schemas supporting complex relationships between customers, service requests, technician profiles, job statuses, and billing records, enabling efficient queries across all modules.
- Built secure REST API endpoints with JWT-based authentication and role-based access control (RBAC) separating Admin, Technician, and Customer permissions — preventing unauthorized data access across all routes.

- ▶ Deployed and configured production infrastructure on AWS EC2: set up Ubuntu server, configured Nginx as a reverse proxy, secured with SSL, and managed Node.js processes using PM2 for zero-downtime restarts.
- ▶ Integrated AWS S3 for scalable media storage — enabling secure upload and retrieval of service reports, job photos, and user documents using pre-signed URLs with time-limited access control.
- ▶ Implemented structured payment workflow APIs to handle service billing, subscription renewal reminders, and payment status tracking, integrating with third-party payment providers.

Vegaa – Real-Time Service Booking Platform Mar 2026 – Present

Tech Stack: Node.js, Express.js, Socket.IO, Redis, PostgreSQL, AWS EC2, Nginx

- ▶ Designed and built a real-time on-demand service booking platform inspired by Uber/Rapido — connecting users with nearby service providers with live status updates and location tracking throughout the session lifecycle.
- ▶ Implemented bidirectional, event-driven communication using Socket.IO — handling booking requests, provider acceptance/rejection, real-time ride/service state transitions (requested → accepted → en-route → completed), and in-app messaging between users and providers.
- ▶ Built live GPS location tracking with continuous coordinate emission from provider to user, enabling real-time map updates and ETA estimation based on distance calculations.
- ▶ Leveraged Redis for high-performance session management, socket state persistence, and pub/sub-based horizontal scaling — ensuring socket events are correctly routed across multiple server instances under concurrent load.
- ▶ Engineered provider matching logic to efficiently query and rank nearby available providers based on geolocation, service type, and availability status using optimized PostgreSQL spatial queries.
- ▶ Deployed the full backend stack on AWS EC2 behind an Nginx reverse proxy with WebSocket upgrade support configured — enabling persistent socket connections in a production environment.

EDUCATION

B.Tech – Computer Science & Engineering • RGUKT IIIT Srikakulam June 2021 – May 2025

- ▶ Rajiv Gandhi University of Knowledge Technologies | CGPA: 8.2 / 10.0
- ▶ Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Software Engineering, OOP

Pre-University Course (PUC / Intermediate) • Board of Intermediate Education, AP Aug 2019 – Apr 2021

- ▶ CGPA: 8.9 / 10.0

CERTIFICATIONS

- ▶ SQL Certification — Great Learning | 2024
- ▶ MongoDB Certification — MongoDB University (Official) | 2023